

# Hiranandani Fortune City — “Arena” Web App

**Remediation Plan** — how each audit finding will be fixed, verified and prevented  
Repository C:\Users\FS165\hfc · branch master · commit 7beebdb · 8 September 2026  
Companion document to *HFC\_Arena\_QA\_Audit\_Report.pdf*

## 1. How this plan is organised

Every issue from the audit appears below with four things: the **root cause**, the **exact code change**, a **verification step** that proves the fix worked, and an **effort estimate**. Issues are grouped into four delivery batches so the work can ship incrementally — Batch 1 alone removes both crashes and can go out the same day.

Where a fix has more than one sensible option, the recommended one is marked and the alternatives are stated so the team can decide. Nothing in this plan changes visual design or content; it is all defect repair, performance and hygiene.

## 2. Delivery batches at a glance

| Batch                   | Contents         | Effort           | Outcome                                                                                                  |
|-------------------------|------------------|------------------|----------------------------------------------------------------------------------------------------------|
| 1 — Hotfix              | F-01, F-02, F-03 | ~2 hours         | No more white-screen crashes; Ebony & Golden Willows panoramas working                                   |
| 2 — Visible UX          | F-04 ... F-10    | ~1 day           | Navbar always present, amenities panel visible, correct icons, single-tap unit opening, no console noise |
| 3 — Performance         | F-11, F-12       | ~2 days          | Heaviest page drops from 43 MB to roughly 3–5 MB; build output shrinks dramatically                      |
| 4 — Hygiene & hardening | F-13 ... F-22    | ~2 days          | Clean install, no security advisories, accessibility & SEO basics, lint-clean codebase, CI gate          |
| Total                   |                  | ≈ 5 working days | One developer, sequential                                                                                |

**Suggested working method.** One branch per batch (fix/batch-1-crashes etc.), each merged after the verification steps in this document pass. Batch 1 should be released on its own — it is low-risk, touches four lines of logic, and removes the two failures a visitor is most likely to hit in a sales lounge.

### 3. Batch 1 — Hotfix (same day)

#### F-01 Gallery "previous" button crashes the app

**File:** src/Pages/Arena/GalleryPage.tsx:310 · **Effort:** 5 min · **Risk:** none

**Root cause.** The handler uses a bare modulo. At `currentIndex === 0` it evaluates to `-1`, and the render body immediately reads `images[currentIndex].title` on an `undefined` element, throwing `TypeError: Cannot read properties of undefined` and unmounting the React tree.

#### CHANGE

```
- onClick={() => setCurrentIndex((prev) => (prev - 1) % images.length)}  
+ onClick={() => setCurrentIndex((prev) => (prev - 1 + images.length) % images.length)}
```

This is the same wrap the file already uses correctly in the lightbox's `onMovePrevRequest` handler, so it is consistent with existing code.

#### HARDENING (RECOMMENDED, SAME COMMIT)

Guard the render against any future out-of-range index so a similar mistake can never blank the screen again:

```
+ const current = images[currentIndex] ?? images[0];  
... then use current.src / current.title instead of images[currentIndex].src / .title
```

**Verify:** open `/arena_gallery`, click ◀ on the first image — it must wrap to the last image with no blank screen and no console error. Repeat on all three tabs (Interior, Elevation, Landscape).

**F-02 Hook-order crash on Pavilion & Olympus unit pages**

**Files:** `src/Towers/Tower3_Pavilion/Unit_Pavilion.tsx:33-53`, `src/Towers/Olympus/Unit_Olympus.tsx:33-55` · **Effort:** 15 min  
 · **Risk:** none

**Root cause.** Both files place the `if (!singleUnit) return ...` guard *before* a `useEffect`. When the same component instance re-renders with a missing unit (browser back/forward between unit URLs, or any navigation that changes only `:id`), React executes fewer hooks than the previous render and tears down the tree: `Rendered fewer hooks than expected.`

**CHANGE — MOVE THE EFFECT ABOVE THE GUARD (BOTH FILES)**

```
const [svgTooltip, setSvgTooltip] = useState<...>(null);

+ useEffect(() => {
+   setSvgTooltip(null);
+   setClickedRoomDefault(null);
+   setClickedRoom2D(null);
+ }, [activeLayout]);

// fallback if unit not found
if (!singleUnit) {
  return ( ... Unit not found ... );
}

- useEffect(() => { ... }, [activeLayout]); // delete from here
```

`Unit_Olympus.tsx` also calls `setHoveredRoom(null)` inside the effect — keep that line when moving it. The other six tower unit pages already declare all hooks before the guard and need no change.

**Verify:** `npx eslint src/Towers` must report zero react-hooks/rules-of-hooks errors. Then open `/arena_unitpavilion/101`, edit the URL to `/arena_unitpavilion/999999` and press browser Back and Forward a few times — the "Unit not found" card must appear each time instead of a blank screen.

**F-03 Ebony & Golden Willows panoramas show a 404 inside the iframe**

**Files:** src/Pages/Ebony/EntrancePageEbony.tsx:14, src/Pages/GoldenWillows/EntrancePageGoldenWillows.tsx:13, public/\_redirects · **Effort:** 30 min – 1 h · **Risk:** low

**Root cause.** Both iframes point at `pano/pano3/index.html` and `pano/pano1/index.html`, but no `pano/` folder exists in `public/`. The SPA fallback rule `/* /index.html 200` answers the missing path with the application itself, so the iframe loads the app inside the app and renders its own 404 screen.

**OPTION A — POINT AT THE HOSTED TOURS (RECOMMENDED)**

The Arena entrance and home pages already use hosted Pano2VR tours and work correctly. Do the same here — no large binaries enter the repo, and tours can be updated without a redeploy:

```
- src="pano/pano3/index.html"
+ src="https://futeservices.com/26-27/Hiranadani/Ebony/index.html"    <-- confirm the real URL with the
360 team
```

**OPTION B — SHIP THE TOURS LOCALLY**

Copy the exported Pano2VR folders into `public/pano/pano1/` and `public/pano/pano3/`, and make the paths root-relative so they resolve from any route:

```
- src="pano/pano3/index.html"    (relative – breaks on nested routes)
+ src="/pano/pano3/index.html"
```

**IN BOTH CASES — STOP THE SPA FALLBACK FROM MASKING MISSING FILES**

Add an exclusion above the catch-all in `public/_redirects` so a genuinely missing tour returns a real 404 instead of silently rendering the app:

```
/pano/*    /pano/:splat    404
/*         /index.html    200
```

**Verify:** load `/ebony` and `/goldenwillows` — the panorama must be interactive inside the frame. Confirm the frame's own URL is the tour, not the app, and that its content is not the "404 — Oops!" page.

**Decision needed:** `/ebony` currently has no link anywhere in the UI — the entrance page sends visitors to the external `ebony.futeservices.in` instead. Confirm whether the in-app Ebony page should be restored (fix it and link it) or deleted along with its route.

## 4. Batch 2 — Visible UX defects (~1 day)

### F-04 Navbar disappears on Gallery and Project Status

**File:** `src/App.tsx:70` and `:75` · **Effort:** 10 min · **Risk:** none

**Root cause.** The navbar renders only for routes listed in `showNavbarPages`, and two of the navbar's own menu items — `/arena_gallery` and `/project_status` — are missing from that list. Confirmed by measurement: those two pages render 0 nav links while every other Arena page renders 7–8.

```
const showNavbarPages = ["/arena", "/arena_projecthighlights",
  "/arena_location", "/arena_master-plan", "/arena_project_details",
+  "/arena_gallery", "/project_status",
  "/arena_pavilion", "/arena_citadel", ... ];

const showNavbarGoldenWillows = ["/goldenwillows", "/golden_projecthighlights",
  "/golden_location", "/project_status_golden", "/projectdetails_golden",
+  "/goldenwillowslayout", "/gallery_golden"];
```

#### PREVENTING THE CLASS OF BUG

The list is maintained by hand and will drift again. Derive it from the menu definition instead — export `menuItems` from `Navbar.tsx` and build the route list from it, so adding a menu item can never again forget to show the navbar:

```
+ const showNavbarPages = [...menuItems.map(i => i.href), ...towerRoutes];
```

**Verify:** click every item in both navbars in turn — the navbar must remain visible on every destination, including Gallery, Project Status, Golden Willows Layout and Golden Gallery.

### F-05 Master-plan amenities panel never renders (CSS typo)

**File:** `src/components/MasterPlanDescription.tsx:5` · **Effort:** 2 min · **Risk:** low — content becomes visible for the first time, so review it

**Root cause.** `lg:bloack` is a misspelling of `lg:block`. Tailwind generates no rule for the invalid class, so the sibling `hidden` is never overridden and the “Tower-wise indoor amenities / Proposed indoor amenities” panel is invisible at every breakpoint.

```
- <div className='absolute top-10 right-0 hidden lg:bloack'>
+ <div className='absolute top-10 right-0 hidden lg:block'>
```

**Verify:** open `/arena_master-plan` at  $\geq 1024$  px width — the panel must appear at the top right. **Then have marketing confirm the listed amenities are still accurate**, since this text has effectively never been shown to visitors. If the content is outdated, either update it or delete the component deliberately rather than leaving it hidden by accident.

**F-06 Gallery play/pause icon is inverted****File:** src/Pages/Arena/GalleryPage.tsx:324 · **Effort:** 2 min · **Risk:** none

**Root cause.** The gallery shows a "play" glyph while the slideshow is running and a "pause" glyph while it is stopped — the opposite of the convention, and the opposite of `PureSlider.tsx:91`, which implements the same control correctly.

```
- {paused ? <MdPause /> : <MdPlayArrow />}
+ {paused ? <MdPlayArrow /> : <MdPause />}
```

While in this block, also add accessible labels to all three controls — they are icon-only buttons with no text:

```
+ aria-label="Previous image" / aria-label={paused ? "Play slideshow" : "Pause slideshow"} / aria-label="Next image"
```

**Verify:** on load the slideshow auto-advances and shows a pause icon; after clicking, it stops and shows a play icon.

**F-07 Opening a unit needs a double-click in 7 of 8 towers****Files:** the seven `Floor_*.tsx` files listed below · **Effort:** 1–2 h including retest · **Risk:** medium — interaction change, test on the kiosk

**Root cause.** Arcadia navigates on `onClick` (`Floor_Arcadia.tsx:131`); Pavilion, Grandstand, Citadel, Greenfield, Athletica, Stadia and Olympus require `onDoubleClick`. On a touchscreen a double-tap is unreliable and undiscoverable, so visitors are likely to conclude the unit plans do not open at all.

**Complication.** In several of those files a single click already does something else — it selects/highlights the unit (`setHoveredUnit` / `setSelectedUnit`). So this is not a blind find-and-replace; the two behaviours must be reconciled.

**RECOMMENDED INTERACTION**

1. First tap on a unit polygon → select it and show its tooltip (current single-click behaviour).
2. Tap on the *already selected* unit → navigate to the unit page.
3. Keep `onDoubleClick` registered as well, so existing muscle memory still works.

```
onClick={e => {
  e.stopPropagation();
+ if (selectedUnit === unit.id) { navigate(`/arena_unitstadia/${unit.id}`); return; }
  setSelectedUnit(unit.id);
}}
onDoubleClick={() => navigate(`/arena_unitstadia/${unit.id}`)}
```

Apply the same pattern in each file, changing only the route prefix. Alternatively, if the tooltip is not important, navigate on the first tap everywhere — simpler, and matches Arcadia. Either way **all eight towers must end up identical**.

**Verify:** on each of the eight towers, open a floor and reach a unit page using only single taps. Confirm on the actual kiosk touchscreen, not just a mouse.

**F-08 Debug logging left in production**

**Files:** Floor\_Arcadia:22, Floor\_Citadel:22, Floor\_Grandstand:22, Floor\_Greenfield:22, Floor\_olympus:24, Floor\_Stadia:24, Floor\_Pavilion:24 · **Effort:** 10 min · **Risk:** none

Delete the seven `console.log("singleFloor", singleFloor)` statements. Each logs an object containing every unit and room on the floor, on every render.

**PREVENTION**

Add a lint rule so this cannot be merged again, and strip any survivors at build time:

```
eslint.config.js  rules: { "no-console": ["error", { allow: ["warn", "error"] }] }
vite.config.ts    esbuild: { drop: ["console", "debugger"] } // production builds only
```

**Verify:** open each of the eight tower floor pages with DevTools open — the console must stay clean.

**F-09 Duplicate React keys in three datasets**

**Files:** src/data/Aminities.ts, src/data/GoldenMasterPlanData.ts, src/Pages/Club\_House/ClubHouse.tsx:43,51 and the components that map them · **Effort:** 1 h · **Risk:** low

**Root cause.** The numeric `id` doubles as the on-screen label *and* as the React key. Repeating a label is legitimate — three separate “Drop Off node” markers genuinely share number 4 — but repeating a key is not. React logs *“Encountered two children with the same key ... the behavior is unsupported and could change in a future version.”*

| Source                           | Duplicate ids                                  | Rendered by                                                       |
|----------------------------------|------------------------------------------------|-------------------------------------------------------------------|
| Aminities.ts                     | 4, 14, 21, 22, 23, 54                          | MasterPlanPage.tsx:129 — <code>&lt;g key={item.id}&gt;</code>     |
| GoldenMasterPlanData.ts          | 1, 3, 4, 20, 21, 29, 43, 46, 47                | GoldenWillowsLayout.tsx:84 — <code>&lt;g key={item.id}&gt;</code> |
| ClubHouse.tsx <code>data1</code> | 3 — used for both “Second Floor” and “Outdoor” | ClubHouse.tsx:79                                                  |

**Current impact, stated precisely.** Nothing is visibly missing today — all 96 master-plan markers render, and the Club House “Outdoor” section does appear. This is a latent correctness risk that React explicitly warns may change behaviour, plus wasted re-rendering. It should be fixed, but it is not an emergency.

**CHANGE — KEEP `ID` FOR DISPLAY, KEY ON SOMETHING UNIQUE**

```
- {amenities.map((item) => { ... return (<g key={item.id}> ... ) }}}
+ {amenities.map((item, i) => { ... return (<g key={`-${item.id}-${item.coords}`}> ... ) }}}
  (or simply key={i} — the arrays are static and never reordered)

ClubHouse.tsx - { id: 3, title: "Outdoor", ... }
               + { id: 4, title: "Outdoor", ... } — a genuine numbering mistake, not an intentional repeat
```

Also remove the redundant inner `key={item.id}` on the nested `<div>` at GoldenWillowsLayout.tsx:51 — a key on a non-list element has no effect.

**Verify:** open /arena\_master-plan, /goldenwillowslayout and /club-house with DevTools — zero key warnings. Count the master-plan markers: still 96.

**F-10 Missing key on the Olympus tooltip list****File:** src/Towers/Olympus/Olympus.tsx:145 · **Effort:** 2 min · **Risk:** none

41 floors are mapped to `<Tooltip>` elements with no key. Stadia.tsx:66 shows the correct form; copy it.

```
{floorsData.map((floor: any) => (  
  <Tooltip  
+   key={`tooltip-${floor.id}`}  
    id={`tooltip-${floor.id}`} ... />  
))}
```

While here, drop the unused `_index` parameters flagged by ESLint at lines 117 and 145, and consider rendering a single shared tooltip driven by `hoveredFloor` instead of 41 mounted components.

**Verify:** /arena\_olympus loads with no "unique key prop" warning; floor tooltips still appear on hover.



## 5. Batch 3 — Performance (~2 days)

### F-11 Page weight — 43.5 MB on the heaviest page

**Scope:** `src/assets/**` and `GalleryPage.tsx` · **Effort:** 1–1.5 days · **Risk:** low, but every affected page needs a visual check

This is the single biggest win available. Measured on the production build with the cache disabled:

| Route                           | Now      | Target         | Main contributor                                        |
|---------------------------------|----------|----------------|---------------------------------------------------------|
| <code>/club-house</code>        | 43.47 MB | ≈ 3 MB         | 16 slider images, one of them 10.81 MB                  |
| <code>/arena_gallery</code>     | 27.94 MB | ≈ 2 MB initial | 21 full-size images loaded at once for 96 px thumbnails |
| <code>/arena_stadia</code>      | 7.47 MB  | ≈ 0.8 MB       | <code>stadia_updated.png</code> — 6.99 MB               |
| <code>/arena_master-plan</code> | 6.83 MB  | ≈ 1.2 MB       | <code>MASTERPLAN-55.png</code> — 6.35 MB                |

#### STEP 1 — RE-ENCODE THE OVERSIZED SOURCE IMAGES

Every asset above ~1.5 MB is re-encoded to WebP with the long edge capped at 2560 px (master plans, which are zoomed, may keep 3840 px). Use `sharp-cli` or `Squoosh`; keep the originals in a separate archive folder outside `src/`.

```
npx @squoosh/cli --webp '{"quality":78}' --resize '{"width":2560}' src/assets/**/*.png
# worst offenders to do first:
exterior5.png 34 MB · Citadel_Lobby_HI-RES.jpg 18 MB · Olympus111.webp 12.8 MB · club4.webp 11.3 MB
Cam01_Grandstand_worm_eye.png 10.2 MB · sports3.webp 10.2 MB · Cam02_Grandstand_close_up.png 9.6 MB
```

Check each converted image side by side with the original at full-screen size before accepting it. These are marketing renders — quality matters, so 78 is a starting point, not a rule.

#### STEP 2 — GIVE THE GALLERY REAL THUMBNAILS

The 96×96 thumbnail strip currently loads the same multi-megabyte files as the main viewer. Generate a 200 px variant per image and reference it in the strip:

```
src/assets/gallery/thumbs/interior1.webp (~15 KB each instead of ~2.9 MB)

const InteriorImages = [
  { src: interior1, thumb: interior1Thumb, title: '2BHK Guest Bedroom' }, ...
];
...
<img src={img.src img.thumb} ... className="w-24 h-24 object-cover ..." />
```

Also load only the current, previous and next full images eagerly; leave the rest to `loading="lazy"` (already set) and drop the eager decode of the whole tab.

#### STEP 3 — LAZY-LOAD THE CLUB HOUSE CAROUSEL

All 16 slider images download on page load even though 15 are off-screen. Render only the active slide plus its neighbours, or add `loading="lazy"` and `decoding="async"` to the mapped `<img>` elements in `ClubHouse.tsx:112`.

**Verify:** re-run the measurement — Chrome DevTools ▶ Network ▶ Disable cache, then read the “transferred” total per route. Targets: club-house under 5 MB, gallery under 3 MB on first paint, no single asset over 2 MB. Compare each page visually against the current build before signing off.

**F-12 One 6.3 MB JavaScript bundle**

**Files:** src/App.tsx, vite.config.ts · **Effort:** 3–4 h · **Risk:** low

**Root cause.** All eight tower datasets — about 122,000 lines of polygon coordinates — are statically imported at the top of App.tsx, so every visitor downloads and parses all of them just to see the entrance page. Vite already warns about this.

```
- import Stadia from './Towers/Stadia/Stadia.tsx'
+ const Stadia = lazy(() => import('./Towers/Stadia/Stadia.tsx'))
  ...same for every tower, floor and unit page, plus Gallery and Club House...

+ <Suspense fallback={<LoadingScreen />}>
  <Routes> ... </Routes>
+ </Suspense>
```

Use a branded loading screen, not a blank div — on a kiosk a flash of white looks like the crash we just fixed. Optionally add a `manualChunks` rule to keep React, MUI and the router in a long-cached vendor chunk.

**Verify:** npm run build emits multiple chunks with no chunk-size warning; the entrance page's initial JS drops well below 500 KB gzipped; navigating to each tower still works with throttling on.

## 6. Batch 4 — Hygiene & hardening (~2 days)

### F-13 npm install fails on a clean checkout

**File:** package.json · **Effort:** 15 min (option A) or 2–3 h (option B) · **Risk:** low / medium

**Root cause.** react-18-image-lightbox@5.1.4 declares `peer react@"16.x || 17.x || 18.x"`; the project runs React 19. npm aborts with ERESOLVE. The library works fine at runtime — this is an install-time blocker for new developers and CI.

#### OPTION A — UNBLOCK INSTALLS NOW (DO THIS IMMEDIATELY)

```
.npmrc (new file, committed)
legacy-peer-deps=true
```

#### OPTION B — REMOVE THE MISMATCH PROPERLY (SCHEDULE SEPARATELY)

Replace the lightbox with a maintained, React-19-compatible one. yet-another-react-lightbox is the closest drop-in; it is used in one place only (GalleryPage.tsx), so the change is contained. Option A stays in place until this lands.

**Verify:** delete node\_modules and package-lock.json, run `npm install` with no flags — it must succeed. Then open /arena\_gallery and confirm the lightbox still opens, navigates and closes.

### F-14 18 npm security advisories (13 high)

**Effort:** 1–2 h including a full regression pass · **Risk:** medium — toolchain upgrade, so retest the build

All affected packages are build-time tooling — vite, rollup, @babel/core, brace-expansion, browserslist, ajv, yaml, @humanfs/node, launch-editor. None ship to the browser, so the live site is not directly exposed; the Vite dev-server advisories do matter for developer machines.

```
npm audit fix
npx update-browserslist-db@latest
npm update react react-dom react-router-dom @mui/material @mui/icons-material react-icons
# then, deliberately, a full regression pass – see section 7
```

Runtime dependencies are also 1–2 minor versions behind (React 19.2.0 → 19.2.8, React Router 7.9.3 → 7.18.3, MUI 7.3.4 → 7.3.11). Take those in the same window so there is one retest, not three.

**Verify:** `npm audit` reports 0 vulnerabilities; `npm run build` succeeds; the regression checklist in section 7 passes.

### F-15 Favicon is a 157 KB PNG served as image/svg+xml

**Files:** index.html:5, public/ · **Effort:** 20 min · **Risk:** none

**Root cause.** public/vite.svg is not an SVG — it is a byte-identical copy of logo2.png (verified by checksum) left over from the Vite template, and the declared MIME type does not match the content.

```
1. Export the Hiranandani mark as a 32/64/180 px PNG set → public/favicon-32.png, favicon-180.png
2. Delete public/vite.svg
- <link rel="icon" type="image/svg+xml" href="/vite.svg" />
+ <link rel="icon" type="image/png" sizes="32x32" href="/favicon-32.png" />
+ <link rel="apple-touch-icon" href="/favicon-180.png" />
```

**Verify:** the correct icon appears in the browser tab and when the kiosk page is added to a home screen.

**F-16 Accessibility & social-sharing basics**

**Files:** all components with `<img>`, `index.html`, `EntrancePage.tsx` · **Effort:** 3–4 h · **Risk:** none

**A) ALT TEXT — 52 OF 61 IMAGES HAVE NONE**

Add descriptive `alt` everywhere. Gallery and unit images already carry a `title` in the data, so the text exists — it just is not being passed through:

```
<img src={img.src} alt={img.title} ... />           gallery + thumbnails
   Header + WithoutbgHeader
<img src={e.image} alt={`Club House view ${e.id}`} />           ClubHouse slider
<img src={bgImage} alt="" />                                   decorative backgrounds: empty alt, deliberately
```

**B) META DESCRIPTION & OPEN GRAPH — LINKS CURRENTLY PREVIEW AS NOTHING**

```
+ <meta name="description" content="Hiranandani Fortune City, Panvel – The Arena. 10.19 acres, 80% open
space, 80+ active-life amenities." />
+ <meta property="og:title" content="The Arena – Hiranandani Fortune City, Panvel" />
+ <meta property="og:description" content="..." />   <meta property="og:image" content="https://.../og-cover.jpg"
/>
+ <meta property="og:type" content="website" />       <meta name="twitter:card" content="summary_large_image" />
```

This matters commercially — the site is shared on WhatsApp, where a missing `og:image` means no preview card at all.

**C) INVALID NESTING — SIX `<A>` INSIDE `<BUTTON>`**

In `EntrancePage.tsx` each entry is `<button><Link>...</Link></button>`, which is invalid HTML and breaks keyboard and screen-reader navigation. Style the link itself and drop the button:

```
- <button className="..."><Link to="/arena">THE ARENA</Link></button>
+ <Link to="/arena" className="...block text-center...">THE ARENA</Link>
```

Use a plain `<a href>` with `target="_blank" rel="noopener noreferrer"` for the four external destinations (Elena, Ebony, Golden Willows).

**Verify:** Lighthouse accessibility score  $\geq 90$ ; `document.querySelectorAll('img:not([alt])').length === 0` on every route; paste the production URL into WhatsApp and confirm the preview card renders.

**F-17** 68 ESLint problems**Effort:** 4–5 h · **Risk:** low

| Count | Rule                                           | Approach                                                                                                                                                                                                                                                                                                                                                                   |
|-------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2     | react-hooks/rules-of-hooks                     | Already fixed in F-02                                                                                                                                                                                                                                                                                                                                                      |
| 40    | no-explicit-any                                | Define <code>Floor</code> , <code>Unit</code> and <code>Room</code> interfaces once in <code>src/types/tower.ts</code> , export them from every data module, and replace <code>floor: any</code> / <code>room: any</code> . All eight datasets share the same shape, so this is one interface reused 40 times.                                                             |
| 14    | no-unused-vars                                 | Delete dead imports and unused state — e.g. <code>isLargeScreen</code> ( <code>Stadia.tsx:14</code> ), <code>setSelectedUnit</code> ( <code>Floor_olympus.tsx:20</code> ), <code>hoveredRoomDefault</code> / <code>hoveredRoom2D</code> ( <code>Unit_Stadia.tsx:24,27</code> ), and the eight stray <code>declare const</code> lines in <code>GalleryPage.tsx:82–86</code> |
| 9     | no-wrapper-object-types                        | Mechanical: <code>useState&lt;Number   null&gt;</code> → <code>useState&lt;number   null&gt;</code>                                                                                                                                                                                                                                                                        |
| 3     | prefer-const, ban-ts-comment, unused directive | <code>npx eslint . --fix</code> handles most; replace <code>@ts-ignore</code> with <code>@ts-expect-error</code> in <code>PureSlider.tsx:25</code>                                                                                                                                                                                                                         |

**Verify:** `npx eslint .` exits clean, and `npx tsc -b` still reports zero errors.**F-18** Slider stalls permanently if Wistia fails to load**File:** `src/components/PureSlider.tsx:22–41` · **Effort:** 30 min · **Risk:** none

**Root cause.** While slide 0 is showing, auto-advance depends entirely on the Wistia JS API binding an `"end"` event. If Wistia is blocked, slow, or the video never fires `end`, no timer is ever set and the Club House carousel stops on the video forever.

```

if (index === 0 && videoRef.current) {
  const readyTimer = setTimeout(handleWistiaReady, 1000);
  + // safety net: advance anyway if the video never reports completion
  + const fallback = setTimeout(() => setIndex(i => (i + 1) % slides.length), 150000);
  + return () => { clearTimeout(readyTimer); clearTimeout(fallback); };
  - return () => clearTimeout(readyTimer);
}

```

150 s matches the interval already configured for slide 0 in `ClubHouse.tsx:93`.**Verify:** block `fast.wistia.net` in DevTools ▶ Network request blocking, load `/club-house`, and confirm the carousel still advances.

**F-19** **Deprecated** `allowTransparency` prop**File:** `src/components/VideoSlide.tsx:7` · **Effort:** 2 min

React 19 no longer recognises this attribute and logs an error on `/club-house`. It has had no effect in modern browsers for years — delete the line.

**Verify:** `/club-house` console is clean of the "React does not recognize the prop" error.

**F-20** **Duplicate images inside the gallery tabs****File:** `src/Pages/Arena/GalleryPage.tsx:97–150` · **Effort:** 30 min · **Risk:** content decision — confirm with marketing

The Interior tab repeats `interior7/8/9` a second time and the Elevation tab repeats `exterior1-6`, so visitors scroll past the same six tower renders twice. Separately, three Olympus elevation images (`exterior13/14/15`) are imported but commented out of the list — they load into the bundle and are never shown.

Remove the repeated entries; decide whether the three Olympus renders should be added back (Olympus is now a live tower, so they probably should) or the imports deleted.

**Verify:** each tab's thumbnail strip contains no visually identical neighbours; new `Set(images.map(i => i.src)).size === images.length`.

**F-21** **Dead code and unreachable routes****Effort:** 1–2 h · **Risk:** low — decisions needed before deleting

- `src/Pages/Arena/GalleryPage1.tsx` (321 lines) — imported nowhere. Delete.
- `src/Pages/GoldenWillows/Towers_Golden/{MainTowerPage,FloorPage,UnitPage}.tsx` (682 lines) — routes commented out at `App.tsx:145–147`. **Decide:** is a Golden Willows tower explorer still planned? If yes, keep and ticket it; if no, delete with the routes.
- `/arena_quality` — registered but its navbar entry is commented out (`Navbar.tsx:13`), so nothing links to it. It duplicates `/quality`, which is richer (5-tab Vimeo player). **Recommend:** delete `/arena_quality` and `Arena_Quality.tsx`, and add "Quality" to the Arena navbar pointing at `/quality`.
- `/ebony` — see the decision noted under F-03.
- Four files carry 230–300 commented-out lines each. Delete them; git history is the archive.

**Verify:** `npm run build` succeeds after deletion; every route in `App.tsx` is reachable by clicking from the entrance page.

**F-22 Repository size — 218 MB of unreferenced assets**

**Effort:** 2–3 h (plus a decision on history rewriting) · **Risk:** medium if history is rewritten — coordinate with the whole team

`src/assets/` is 904 MB, of which **218 MB across 118 files is referenced by no code at all** — superseded master-plan revisions (`arena_masterplan1–4`, `MASTERPLAN-5.png`, `masterplan_new_18_7_2026.webp`), an unused `gallery1/` folder, and old tower renders. The git pack is 772 MB, so a fresh clone is painfully slow.

1. Archive the 118 unreferenced files somewhere outside the repo, then delete them from `src/assets/`. A re-run of the unused-asset scan produces the exact list.
2. Adopt Git LFS for `*.png *.jpg *.jpeg *.webp` going forward, so new renders do not bloat the pack.
3. Optionally rewrite history with `git filter-repo` to reclaim the 772 MB. **This rewrites every commit hash** — it needs a scheduled window and everyone re-cloning. Treat as a separate task, not part of this remediation.
4. Delete the stray `Thumbs.db` files and add `Thumbs.db / .DS_Store` to `.gitignore`.

**Verify:** the unused-asset scan reports 0 unreferenced files; `npm run build` succeeds; every page still renders every image (the route sweep in section 7 covers this).

## 7. Regression checklist — run after every batch

These are the checks that caught the defects in the first place. They are quick and should be repeated before each merge.

| Check         | How                                                                                       | Pass condition                                       |
|---------------|-------------------------------------------------------------------------------------------|------------------------------------------------------|
| Clean install | <code>rm -rf node_modules package-lock.json</code><br><code>&amp;&amp; npm install</code> | Succeeds with no flags                               |
| Types         | <code>npx tsc -b</code>                                                                   | 0 errors                                             |
| Lint          | <code>npx eslint .</code>                                                                 | 0 errors (0 warnings after Batch 4)                  |
| Build         | <code>npm run build</code>                                                                | Succeeds; no chunk-size warning after F-12           |
| Route sweep   | Visit all 29 static routes + one floor and one unit per tower                             | Every page renders; console clean; no broken images  |
| Crash paths   | Gallery ◀ on first image; unit page valid → invalid id via Back/Forward                   | No blank screen                                      |
| Embeds        | Open every page containing an iframe                                                      | All 13 frames show real content, never the app's 404 |
| Weight        | DevTools ▶ Network ▶ Disable cache, read "transferred"                                    | No route above 5 MB after Batch 3                    |
| Mobile        | DevTools device toolbar at 390×844                                                        | No horizontal scrollbar; navbar reachable            |
| Kiosk         | The actual touchscreen, full-screen                                                       | Every unit reachable with single taps                |
| Security      | <code>npm audit</code>                                                                    | 0 vulnerabilities after F-14                         |

## 8. Preventing recurrence

Four small changes would have caught most of what this audit found, before it reached the branch:

1. **Make lint a merge gate.** ESLint already detects the hook-order crash (F-02) and every unused variable. A GitHub Action running `npm ci && npx tsc -b && npx eslint . && npm run build` on each pull request turns those into blocking failures.
2. **Ban console in the lint config** and add `esbuild.drop` for production builds (F-08).
3. **Add an image budget.** A pre-commit hook rejecting any image over ~2 MB in `src/assets/` would have prevented the 34 MB and 18 MB files outright.
4. **Derive route lists from data, not by hand.** The navbar gap (F-04) exists purely because two lists must be kept in sync manually.

## 9. Open decisions for the client

| Item         | Question                                                                                         | Blocks     |
|--------------|--------------------------------------------------------------------------------------------------|------------|
| Ebony page   | Restore the in-app Ebony panorama, or delete the page and keep the external link?                | F-03, F-21 |
| Pano hosting | Are hosted tour URLs available for Ebony and Golden Willows, or should the folders be committed? | F-03       |



|                       |                                                                                          |      |
|-----------------------|------------------------------------------------------------------------------------------|------|
| Amenities panel       | The hidden master-plan amenities text — still accurate, or should it be updated/removed? | F-05 |
| Unit interaction      | First tap selects then second opens, or first tap opens directly?                        | F-07 |
| Golden Willows towers | Is the tower explorer still planned, or can those three files go?                        | F-21 |
| Olympus gallery       | Should the three commented-out Olympus elevation renders be shown?                       | F-20 |
| Git history           | Is a scheduled history rewrite acceptable to reclaim ~770 MB?                            | F-22 |

Every finding referenced here was reproduced against the current master build during the audit on 8 September 2026 — the two crashes were captured as live browser exceptions, the broken panoramas were confirmed by reading the iframe’s own document, and all page weights were measured on the production build with the cache disabled. Line numbers refer to commit 7beebdb and will shift as fixes land.